1. Let $v_1 = (1,-2,3)$ and $v_2 = (0,5,-3)$. Verify that the set $\{v_1, v_2\}$ is linearly independent and extend it to a basis of $\mathbb{R}^3$.

2. Let $v_1 = (1,0,0,0)$ and $v_2 = (1,1,0,0)$. Verify that the set $\{v_1, v_2\}$ is linearly independent and extend it to a basis of $\mathbb{R}^4$.

3. Let $v_1 = (1,0,0),\ v_2 = (1,0,1),\ v_3 = (2,0,1),\ v_4 = (0,0,-1)$. Find a basis of the subspace $\text{span}\{v_1, v_2, v_3, v_4\}$ of $\mathbb{R}^3$.

4. Let $v_1 = (1,1,1,1),\ v_2 = (2,2,2,0),\ v_3 = (0,0,0,3),\ v_4 = (3,3,3,4)$. Find a basis of the subspace $\text{span}\{v_1, v_2, v_3, v_4\}$ of $\mathbb{R}^4$.

5. Let $V = \mathbb{R}^{\mathbb{R}}$, the vector space of all functions $f : \mathbb{R} \to \mathbb{R}$. Prove that for every positive integer n we can find $n+1$ linearly independent functions in V (*hint*: look for polynomials). Explain why it follows that $\mathbb{R}^{\mathbb{R}}$ is infinite-dimensional.

6. Determine whether each of the following statements is true or false, and justify your answer.

   (a) $\dim\{0\} = 0$.

   (b) There is a set of 17 linearly independent vectors in F^{17}.

   (c) There is a set of 11 vectors that span F^{17}.

   (d) Every linearly independent set of five vectors in F^5 is a basis of F^5.

   (e) Every set of five vectors that spans F^5 is a basis of $\mathbb{R}^5$.

   (f) There are at least two distinct three-dimensional subspaces of $\mathscr{P}_2(F)$ (the vector space of polynomials of degree at most 2).